discharge tube is kept in between pole pieces of magnet such that both electric and magnetic fields are acting perpendicular to each other. When the cathode rays strike the screen, they produce scintillation and hence bright spot is observed. This is achieved by coating the screen with zinc sulphide.

(i) Determination of velocity of cathode rays

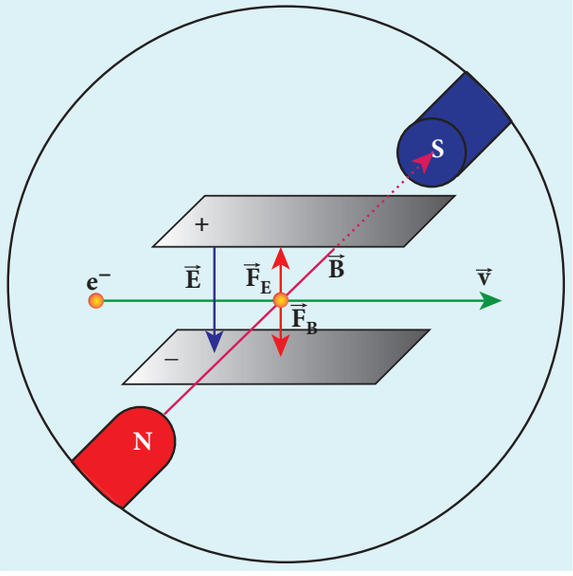


Figure 9.4 Electric force balancing the magnetic force – the path of electron beam is a straight line

For a fixed electric field between the plates, the magnetic field is adjusted such that the cathode rays (electron beam) strike at the original position O (Figure 9.3). This means that the magnitude of electric force is balanced by the magnitude of force due to magnetic field as shown in Figure 9.4. Let e be the charge of the cathode rays, then

$$eE = eB\upsilon$$

$$\Rightarrow \upsilon = \frac{E}{B} \qquad (9.1)$$

(ii) Determination of specific charge

Since the cathode rays (electron beam) are accelerated from cathode to anode, the potential energy of the electron beam at the cathode is converted into kinetic energy of the electron beam at the anode. Let V be the potential difference between anode and cathode, then the potential energy is eV. Then from law of conservation of energy,

$$eV = \frac{1}{2}m\upsilon^2 \Rightarrow \frac{e}{m} = \frac{\upsilon^2}{2V}$$

Substituting the value of velocity from equation (9.1), we get

$$\frac{e}{m} = \frac{1}{2V}\frac{E^2}{B^2} \qquad (9.2)$$

Substituting the values of E, B and V, the specific charge can be determined as

$$\frac{e}{m} = 1.7\times10^{11}\,\mathrm{C\,kg^{-1}}$$

(iii) Deflection of charge only due to uniform electric field

When the magnetic field is turned off, the deflection is only due to electric field. The deflection in vertical direction is due to the electric force.

$$F_e = eE \qquad (9.3)$$

Let m be the mass of the electron and by applying Newton's second law of motion, acceleration of the electron is

$$a_e = \frac{1}{m}F_e \qquad (9.4)$$

Substituting equation (9.4) in equation (9.3),

$$a_e = \frac{1}{m}eE = \frac{e}{m}E$$

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